Status	Finished
Started	Sunday, 11 October 2026, 10:47 AM
Completed	Sunday, 11 October 2026, 10:50 AM
Duration	3 mins 19 secs
Grade	**13.00** out of 15.00 (**86.67**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. Requirement of sorted array is expensive when a lot of insertion and deletions are needed

b. Binary search algorithm is not efficient when the data elements more than 1500 ✓

c. There must be a mechanism to access middle element directly

d. Must use a sorted array

Question **2**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list is unsorted

b. When the list is sorted ✓

c. When the list contains duplicate elements

d. When the list is very small

Question **3**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. An error is raised

b. The first element is returned

c. The search is considered unsuccessful ✓

d. The last element is returned

Question **4**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

- a. When the list is large and unsorted
- b. When the list is small and unsorted ✓
- c. When the list is large and sorted
- d. When the list is small and sorted

Question **5**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

- a. In the entire list
- b. At the beginning of the list
- c. In the left sublist ✓
- d. In the right sublist

Question **6**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

- a. The search backtracks to find duplicate elements
- b. The search stops immediately ✓
- c. The search continues until the end of the list
- d. The search starts over from the beginning

Question **7**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

- a. By dividing the list into halves
- b. By comparing each element in the list sequentially ✓
- c. By using a hash function
- d. By sorting the list first

Question **8**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

a. Discovering
b. Finding
c. Searching ✓
d. Mining

Question **9**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search is more efficient than binary search on large lists.
b. Linear search can be applied to both sorted and unsorted lists. ✓
c. Linear search divides the list into halves.
d. Linear search requires the list to be sorted.

Question **10**

Incorrect

Mark 0.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search has a lower time complexity on large, sorted lists
b. Binary search does not require dividing the list ✗
c. Binary search can find multiple instances of the target element
d. Binary search works on unsorted lists

Question **11**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

a. O(log n)
b. O👎
c. O(1) ✓
d. O(n log n)

Question **12**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

a. Sorting the list before searching
b. Skipping every second element
c. Dividing the list in half repeatedly
d. Checking each element sequentially ✓

Question **13**

Incorrect

Mark 0.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a. 89 and 94
b. 89 and 99
c. 90 and 94 ✗
d. 90 and 99

Question **14**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

a. It compares elements sequentially
b. It can be applied only if the list is sorted ✓
c. It always starts from the beginning of the list
d. It works on unsorted lists

Question **15**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

a. 89 and 94
b. 90 and 100
c. 94 and 99
d. 90 and 99 ✓